Dealer Website & Schedule Automation System

Proof of Concept (POC)

Problem Statement

The organization has an Excel file containing **627 dealer names**. The objective is to automatically identify each dealer's official website, extract their business scheduling/working hours, and store the information in a structured format. Manual searching is time-consuming, error-prone, and not scalable for large datasets.

Proposed Solution

Develop an AI-powered automation pipeline that:

1. Reads dealer names from Excel.
2. Searches for the dealer's official website.
3. Checks whether an API is available.
4. If API exists, extracts scheduling information through the API.
5. If API is unavailable, performs web scraping to collect scheduling details.
6. Uses an LLM (GPT/Gemini) to validate and standardize extracted schedules.
7. Updates the output Excel file with website and scheduling information.

Approach

Workflow

Technologies

- Python
- Pandas
- OpenPyXL
- Google Search API / SerpAPI
- BeautifulSoup
- Selenium / Playwright
- OpenAI GPT / Google Gemini
- APScheduler (for future automation)

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Sample Input

Dealer Name
ABC Motors
XYZ Cars
PQR Auto

Sample Output

Dealer Name	Website	Schedule	Source
ABC Motors	abcmotors.com	Mon-Fri 9AM-6PM	API
XYZ Cars	xyzcars.com	Mon-Sat 10AM-7PM	Scraping

Success Criteria

- Website Identification Accuracy: ≥ 90%
- Schedule Extraction Accuracy: ≥ 80%
- Processing Time: < 30 seconds per dealer
- Manual Review Cases: < 20%

Expected Outcome

The POC demonstrates that dealer website discovery and schedule extraction can be automated efficiently. Once validated on a sample set (20-50 dealers), the solution can be scaled to process all **627 dealer records**, significantly reducing manual effort and improving operational efficiency.

Project Goal: Automate dealer website identification and schedule extraction using a hybrid approach combining Search APIs, Web Scraping, and AI-based validation.